

PROJECT WORK-1(23ET6PWPJ1)
based report
ON
“Design and power optimization of True dual port SRAM”

Submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING
IN
ELECTRONICS AND TELECOMMUNICATION ENGINEERING



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(Autonomous Institution under VTU, Belagavi)

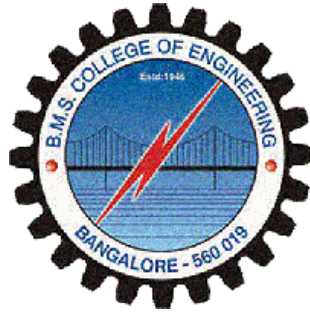
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BONAFIDE CERTIFICATE

This is to certify that the Project work-1(23ET6PWPJ1) entitled “**Design and power optimization of True dual port SRAM**” is submitted by **Arunansh Arya(1BM23ET007)**, **M. Naveen Krishnan (1BM23ET028)**, **Sundaresh A. M. (1BM23ET054)** and **Sindhu R. G. (1BM24ET404)**, in partial fulfillment for the award of degree of Bachelor of Engineering in Electronics and Telecommunication Engineering during the academic year 2025-2026.

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ABSTRACT

Power dissipation in on-chip memory arrays is one of the most critical challenges in modern VLSI design, with SRAM blocks contributing a significant fraction of total chip power in processors, networking ASICs, and embedded systems. Conventional SRAM designs operating at nominal supply voltages exhibit high dynamic power due to continuous wordline and bitline switching, even for rows that are not being accessed in a given clock cycle.

This project presents the design and power optimization of a 12-transistor (12T) True Dual Port (TDP) SRAM in the gpdk090 (90nm) CMOS process at a supply voltage of 1.2V. The 12T topology provides two fully independent read and write ports (Port A and Port B) that operate simultaneously without interference, while a dedicated decoupled read path eliminates the read disturbance problem inherent in standard 6T and 8T cells.

The full design hierarchy from the 12T bit cell through SRAM_Row_16, SRAM_256x16, and SRAM_1024x16 was implemented in Cadence Virtuoso and simulated using the Spectre transient engine. Baseline power measurement of the 1024×16 array (16,384 cells, 196,608 transistors) at 25 MHz yielded 78 mW under worst-case conditions.

Two power optimization techniques were applied: Wordline Segmentation, which divides the 1024-row array into four independent 256-row segments, and Clock Gating, which uses 2 input AND gate based CLOCK GATE cells to block wordline switching in inactive segments. With three of four segments disabled, the optimized design achieved a measured power of 9.619 mW a reduction of approximately 87% compared to the Baseline. These results confirm that wordline segmentation combined with clock gating is highly effective for reducing dynamic power in large SRAM arrays.

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CHAPTER 1

Introduction

The demand for high-performance, power-efficient on-chip memory has grown rapidly with the proliferation of multi-core processors, networking chips, and embedded SoCs. Static Random Access Memory (SRAM) is the dominant choice for on-chip cache and register file implementations due to its speed and density advantages. However, as array sizes increase to thousands of rows and columns, power dissipation in SRAM becomes a significant design constraint.

This project addresses the design and power optimization of a 12-transistor (12T) True Dual Port SRAM in the gpdk090 90nm CMOS process. The 12T cell improves upon standard 6T designs by providing two fully independent ports for simultaneous read and write operations, along with a decoupled read path that eliminates read disturbance. Power optimization is achieved through wordline segmentation and clock gating, together yielding power reduction compared to the unoptimized Baseline.

1.1 Objectives

- To design a 12T True Dual Port SRAM bit cell with fully decoupled read path in gpdk090 (90nm) technology.
- To verify simultaneous and independent dual-port read and write operation through transient simulation.
- To implement wordline segmentation by dividing the 1024-row array into four independent 256-row segments.
- To implement clock gating using 2 input AND based CLOCK GATE cells to block switching in inactive segments.
- To measure and compare Baseline and optimized power dissipation using Cadence Virtuoso and Spectre.

1.2 Scope

The scope of this project involves the design of a 12T True Dual Port SRAM bit cell and its hierarchical scaling to a 1024×16 array. The design includes functional verification of read and write operations on both ports simultaneously. Power optimization is addressed through wordline segmentation and clock gating at the array level. The entire design flow of schematic entry, symbol creation, testbench setup, and transient simulations carried out in Cadence Virtuoso using the Spectre simulator with the gpdk090 process design kit at a 1.2V supply.

1.3 Proposed System

The proposed design consists of a hierarchical 12T True Dual Port SRAM implemented and optimized in Cadence Virtuoso. At the cell level, a 12-transistor topology provides two completely independent ports (Port A and Port B) each capable of simultaneous read and write. The read path uses dedicated buffer transistors that sense the stored node without directly connecting to it, preventing read disturbance entirely.

At the array level, the 1024×16 configuration is organized into four segments of 256 rows each. A 2 input AND based CLOCK GATE cell at the input of each segment's wordlines gates switching activity based on a segment enable signal (SEG_EN). When a segment is disabled, its wordlines are permanently held at logic zero, eliminating all dynamic switching power for that segment. The combination of wordline segmentation and clock gating reduces measured, achieving power reduction.

CHAPTER 2

Literature Survey

This chapter surveys existing research on SRAM cell topologies, dual-port memory design, and power optimization techniques relevant to this project. The review covers foundational work on multi-port SRAM cells, wordline segmentation, and clock gating, providing the theoretical basis for the design decisions made in this project.

2.1 Literature Review

[1] Y. Long, Z. Chen, and H. Wang, "Low-Power SRAM Design Using Optimized Bitline and Wordline Techniques," IEEE access, 2023.

This paper focuses on reducing dynamic power consumption in SRAM arrays by optimizing bitline and wordline switching activity. The authors propose techniques to minimize unnecessary transitions in these lines, which are major contributors to dynamic power. Their work highlights that reducing capacitive switching directly improves energy efficiency, forming a strong basis for incorporating optimized control signals in low-power SRAM design.

[2] H. Kim, S. Lee, and J. Park, "Energy-Efficient Dual-Port SRAM Architecture for High-Performance Systems," IEEE Transactions on VLSI Systems, 2023.

This paper proposes an energy-efficient dual-port SRAM architecture that enables simultaneous read and write operations. The authors analyze power-performance trade-offs and demonstrate that proper architectural design can significantly enhance performance while maintaining low power consumption. Their work supports the implementation of true dual-port SRAM structures in applications requiring parallel memory access.

[3] L. Wang, X. Zhou, and Q. Liu, "Dynamic Power Reduction in SRAM Arrays Using Clock Gating Techniques," IEEE Access, 2024.

This work demonstrates the effectiveness of clock gating in reducing dynamic power consumption in SRAM arrays. By disabling clock signals to inactive portions of the circuit, unnecessary switching activity is avoided. The authors show that this technique significantly lowers power usage without affecting performance, which directly motivates the use of gated control signals in this project.

[4] D. Lee, H. Park, and K. Cho, "Wordline Segmentation for Low-Power Large-Scale SRAM Arrays," IEEE Transactions on Circuits and Systems, 2024.

This paper introduces wordline segmentation as a method to reduce capacitive loading in large SRAM arrays. By dividing long wordlines into smaller segments, only the required segment is activated during operation, thereby reducing switching power. The technique improves overall power efficiency and scalability, making it highly suitable for large memory designs.

[5] R. Patel, A. Sharma, and P. Gupta, "Combined Low-Power Techniques for SRAM in Deep Submicron Technologies," Microelectronics Journal, 2024.

This paper presents a combination of low-power techniques such as clock gating and wordline segmentation to achieve significant power reduction in SRAM designs. The authors demonstrate that integrating multiple optimization strategies results in better performance compared to using individual techniques alone. Their findings support the adoption of hybrid approaches for efficient SRAM design in advanced technologies.

2.2 Comparative Study

Design Type	Ports	Read Disturb	Power Opt.	Complexity	Remarks
Standard 6T SRAM	1 (R/W)	Yes — disturbs Q	None	Low	Widely used; read disturbance limits stability
8T SRAM	1R + 1W	No — decoupled	Limited	Medium	Decoupled read; single port only
10T SRAM	2 ports, limited	Partial	Limited	Medium	Some implementations have port conflicts
12T TDP SRAM (This Work)	2 full R/W	No — fully decoupled	Clock Gate + WL Seg.	High	Power reduction vs Baseline

Table 2.1 – Comparative Analysis of SRAM Design Approaches

2.3 Summary of Findings

From the literature survey:

- Decoupled read paths in multi-transistor SRAM cells effectively eliminate read disturbance and improve SNM.
- Wordline segmentation reduces dynamic power proportionally to the number of segments by limiting per-access switching capacitance.
- Clock gating using AND-gate structures is a proven, low-overhead technique for preventing unnecessary switching in inactive circuit blocks.
- True Dual Port SRAM is essential for multi-core and multi-processor systems where simultaneous independent memory access is required.
- Combining wordline segmentation and clock gating can yield power reductions significantly beyond what either technique achieves alone.

CHAPTER 3

Problem Statement & Motivation

3.1 Problem Statement

Modern VLSI chips contain millions of SRAM cells organized into large arrays that serve as cache memory, register files, and data buffers. As array dimensions grow, power dissipation in SRAM becomes a dominant contributor to overall chip power budget. In a standard unoptimized 1024-row SRAM array operating at 25 MHz, all 1024 wordlines switch every clock cycle regardless of which row is actually being accessed. This wasteful switching activity results in large dynamic power dissipation that scales with array size, clock frequency, and supply voltage.

Simultaneously, conventional 6T SRAM cells support only a single read/write port, which creates a bottleneck in multi-core processor designs where multiple cores may need to access shared memory simultaneously. Dual-port solutions based on 8T cells provide limited improvement but suffer from port conflicts and incomplete independence between the two ports.

Additionally, standard 6T cells suffer from read disturbance — the act of reading can disturb the stored value when the bitline is directly connected to the storage node. This limits operation at low supply voltages and high temperatures. There is a clear need for a SRAM design that provides true simultaneous dual-port operation, disturbance-free reading, and significantly reduced power dissipation through systematic optimization techniques.

3.2 Motivation

The motivation for this project arises from the increasing importance of on-chip memory power in advanced VLSI designs. As processor architectures adopt deeper cache hierarchies and wider datapaths, SRAM array area and power grow proportionally.

Industry reports indicate that cache SRAM can account for 30–50% of total chip power in modern processors, making its optimization a critical design objective. From a research perspective, the 90nm gpd090 process provides an ideal academic vehicle for studying SRAM power trade-offs at a near-industry technology node.

The availability of Cadence Virtuoso and Spectre in academic settings enables full transient simulation with accurate BSIM3v3 transistor models, allowing realistic power measurement without requiring physical fabrication.

The motivation is further reinforced by the fact that two independently validated power reduction techniques wordline segmentation and clock gating are well established in literature but rarely combined and demonstrated together on a True Dual Port cell in an academic design flow. This project bridges that gap by implementing, simulating, and quantifying the combined effect of both techniques on a complete 1024×16 12T True Dual Port SRAM array.

3.3 Existing System

Current SRAM designs used in standard cell libraries and academic benchmarks typically employ 6T cells with a single read/write port. While compact and well-characterized, 6T cells suffer from the read disturbance problem during a read operation, the bitline is directly connected to the storage node Q, allowing the charging bitline to potentially flip the stored bit. This issue becomes more pronounced at reduced supply voltages and higher temperatures.

Existing dual-port SRAM implementations in literature use 8T or 10T topologies that provide limited independence between ports. In many such designs, simultaneous same-address access can cause data corruption or undefined behavior. Furthermore, existing designs typically lack comprehensive power optimization wordline segmentation and clock gating are implemented separately in commercial products but are rarely combined in academic demonstrations with measured results.

CHAPTER 4

System Requirement Specifications

4.1 Functional Requirements

The functional requirements define the essential operations that the 12T True Dual Port SRAM must satisfy to ensure correct functionality and effective power optimization. The primary requirement is simultaneous dual-port operation, where Port A and Port B perform independent read and write operations without interference, data corruption, or signal degradation. This is ensured by electrically isolating the write access and read path transistors of each port. The design must also provide disturbance-free read operation, such that reading does not affect the stored data at nodes Q and QB. This is achieved using a decoupled read path, where the read transistors sense the stored node through gate control without drawing current. For power optimization, the system must implement wordline segmentation to activate only the required segment during operation, and clock gating to disable switching in inactive segments by holding their wordlines at logic zero. These techniques should result in a significant reduction in power compared to the Baseline design.

4.2 Design Specifications

4.2.1 12T Bit Cell Specifications

Transistor	Type	W	L	Function
M1 — Inv 1 PMOS	PMOS	1.0 μm	100 nm	Pull-up left — drives node Q HIGH
M2 — Inv 2 PMOS	PMOS	1.0 μm	100 nm	Pull-up right — drives node QB HIGH
M3 — Inv 1 NMOS	NMOS	500 nm	100 nm	Pull-down left — drives node Q LOW
M4 — Inv 2 NMOS	NMOS	500 nm	100 nm	Pull-down right — drives node QB LOW
M5 — Port A Write L	NMOS	600 nm	100 nm	Connects WBL_A pin to node Q during write
M6 — Port A Write R	NMOS	600 nm	100 nm	Connects WBLB_A pin to node QB during write
M7 — Port B Write L	NMOS	600 nm	100 nm	Connects WBL_B pin to node Q during write
M8 — Port B Write R	NMOS	600 nm	100 nm	Connects WBLB_B pin to node QB during write
M9 — Port A Read Buf	NMOS	400 nm	100 nm	Senses Q for Port A — gate only
M10 — Port A Read Acc	NMOS	400 nm	100 nm	Connects read path to RBL_A pin when RWL_A HIGH
M11 — Port B Read Buf	NMOS	400 nm	100 nm	Senses Q for Port B — gate only
M12 — Port B Read Acc	NMOS	400 nm	100 nm	Connects read path to RBL_B pin when RWL_B HIGH

Table 4.1 – 12T Bit Cell Transistor Specifications

4.2.2 Array Specifications

- Process Technology: gpdk090 (90nm CMOS)
- Supply Voltage: $VDD = 1.2V$
- Array Size: 1024 rows $\times$ 16 columns = 16,384 cells
- Total Transistors: 196,608 ($16,384 \times 12$)
- Clock Frequency: 25 MHz (40ns period)
- Number of Segments (optimized): 4×256 rows
- CLOCK GATE Instances: 16 (4 per segment $\times$ 4 wordline types)

4.2.3 CLOCK GATE Cell

The CLOCK GATE cell implements the logic: $GWL = EN \text{ AND } WL$, using one 2 input AND gates. The three inputs are EN (segment enable from SEG_EN bus), and WL (wordline input — WWL_A, WWL_B, RWL_A, or RWL_B). The output GWL is the gated wordline that drives the SRAM_256x16 segment. When $EN = 0$, $GWL = 0$ regardless of WL, ensuring zero switching activity in the disabled segment.

4.3 Software Requirements

4.3.1 Cadence Virtuoso -Schematic and Design Entry

Cadence Virtuoso Schematic Editor L is the primary design entry tool used for this project. All schematics from the 12T bit cell through the CLOCK GATE cell and the full optimized design are created in Virtuoso using the gpdk090 PDK transistor models. The hierarchical design flow uses automatically generated symbols from schematics, enabling modular assembly of larger blocks.

4.3.2 Cadence Spectre Simulator

The Spectre analog simulator, accessed through Cadence ADE L (Analog Design Environment L), is used for all transient simulations. Spectre uses BSIM3v3 transistor models from the gpdk090 process file (gpdk090.scs, section tt for typical-typical corner) to accurately compute node voltages and branch currents. Transient analyses are run for 200 ns with a time step of 10 ps. Power is measured using the expression $\text{average}(1.2 * IT("/VDD_source/MINUS"))$, which computes total power as the time-average of the product of supply voltage and VDD terminal current.

CHAPTER 5

Methodology

5.1 Block Diagram

The overall design hierarchy and signal flow for the optimized design is illustrated in the block diagram

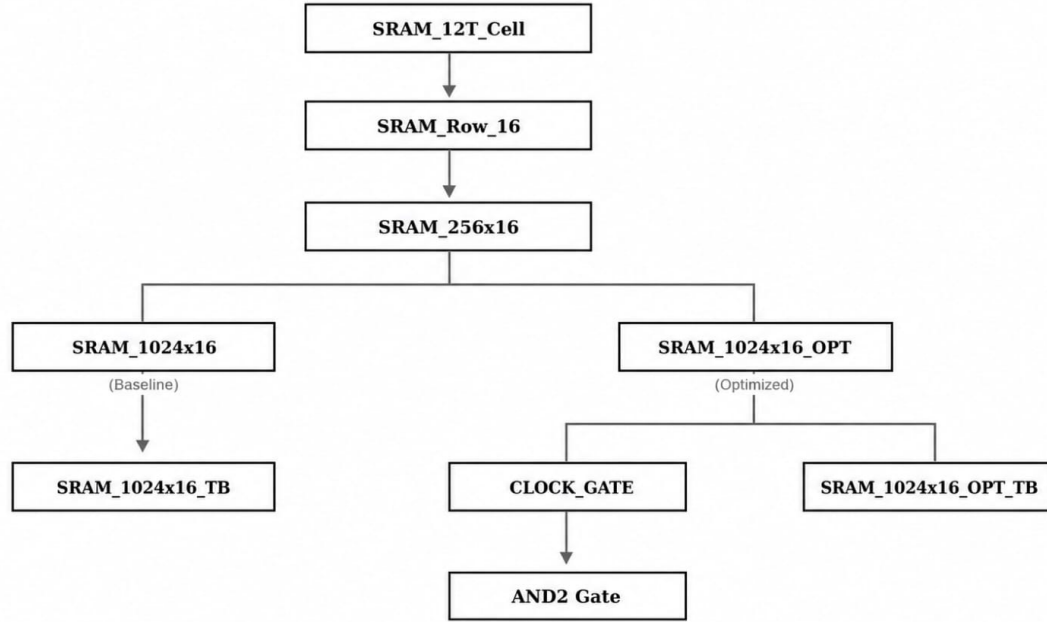


Fig 5.1 – Design Hierarchy Block Diagram

The block diagram shows the hierarchical organization of the optimized SRAM design. At the lowest level, 12T bit cells are organized into rows of 16 cells to form SRAM_Row_16 blocks. Four stacks of 256 such rows constitute each SRAM_256x16 segment. Four such segments together form the complete 1024×16 array. At the optimization layer, each segment's wordline inputs pass through CLOCK GATE cells before reaching the SRAM_256x16 instances. The SEG_EN<3:0> bus and WL signal control the CLOCK GATE cells, enabling or blocking wordline activity per segment. All four segments share common bitline buses and power rails.

5.2 Module Design

• 12T Bit Cell Module

The 12T bit cell consists of three functional parts. The storage core is formed by two cross-coupled CMOS inverters: Inverter 1 (M1 PMOS + M3 NMOS) and Inverter 2 (M2 PMOS + M4 NMOS). These cross-coupled inverters create a positive feedback latch that holds the stored bit indefinitely. The write access part consists of four NMOS transistors: M5 and M6 controlled by WWL_A form Port A's write access, while M7 and M8 controlled by WWL_B form Port B's write access. During a write, the activated wordline turns on the access transistors, allowing the write bitlines to drive Q and QB to the desired logic levels. The read path consists of M9/M10 for Port A and M11/M12 for Port B. M9's gate is connected to Q, when Q = 1, M9 turns on and creates a discharge path from RBL_A through M10 to ground. Since M9 connects to Q only through its gate terminal, no current flows from the storage node during reading, completely eliminating read disturbance.

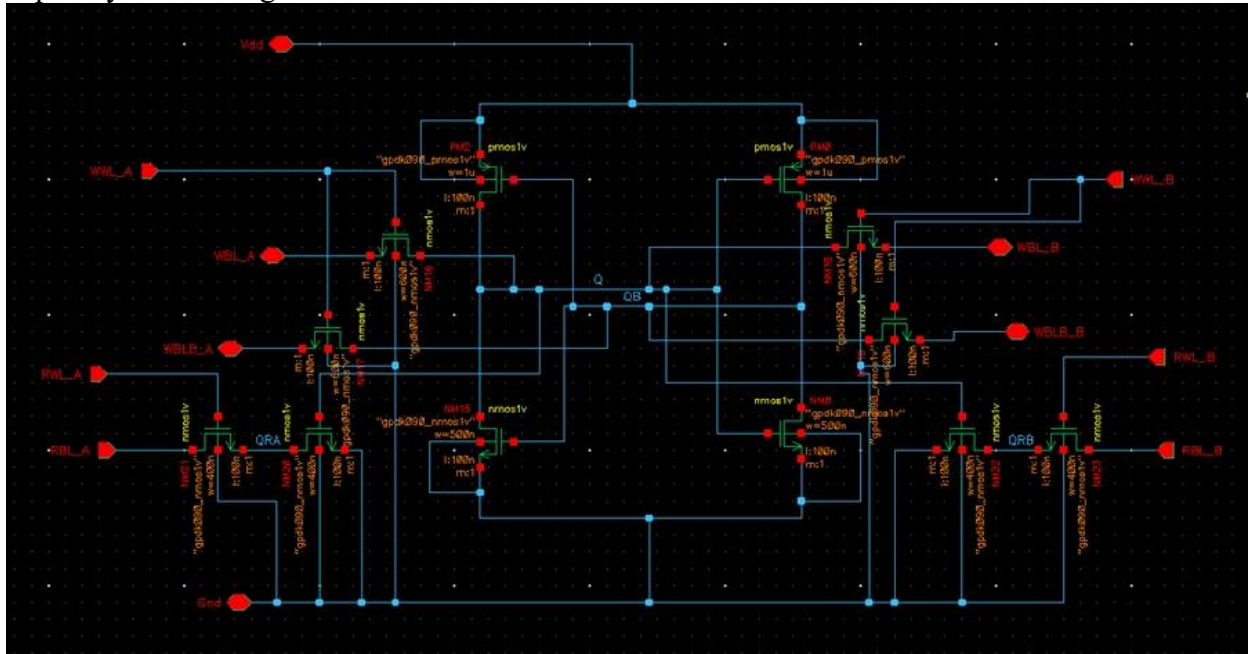


Fig 5.2 :12T SRAM Bit Cell Schematic

• SRAM Array Hierarchy Module

The design hierarchy progresses from SRAM_Row_16 (1 row $\times$ 16 columns) to SRAM_256x16 (256 rows $\times$ 16 columns) and finally SRAM_1024x16 (1024 rows $\times$ 16 columns). Bus notation is used throughout to manage the large number of connections. Each row shares a single set of four wordlines (WWL_A, WWL_B, RWL_A, RWL_B), while each column has its own set of bitlines. Array instancing in Cadence Virtuoso is used to place the required number of row or cell instances efficiently without manual placement.

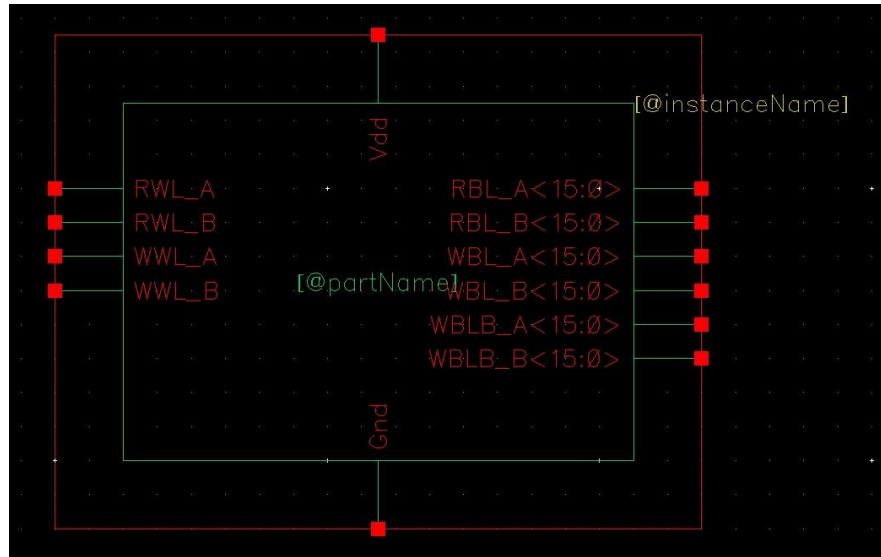


Fig5.3: Symbol of 1 row with 16 bits of data

The SRAM_Row_16 symbol represents a single row of 16 12T SRAM cells, implementing 16 bits of storage width. The symbol exposes write wordlines (WWL_A, WWL_B), read wordlines (RWL_A, RWL_B), write bitlines (WBL, WBLB) and read bitlines (RBL) for both ports A and B, along with power supply pins VDD and GND.

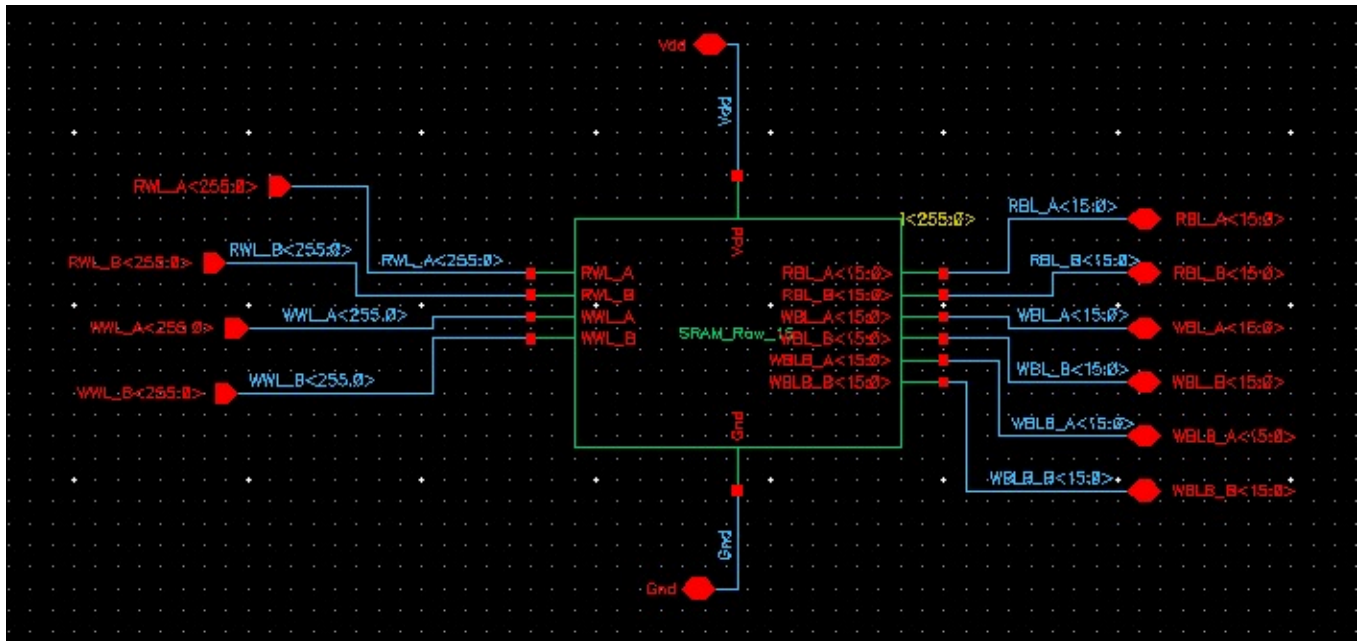


Fig 5.4: Design of 256x16 for wordline segmentation

The SRAM_256x16 array is constructed by instantiating 256 instances of the SRAM_Row_16 cell using Cadence array instantiation, resulting in a 256 row by 16 column memory array with a total capacity of 4096 bits. All wordlines are individually addressable while bitlines are shared across all 256 rows, following standard SRAM array architecture.

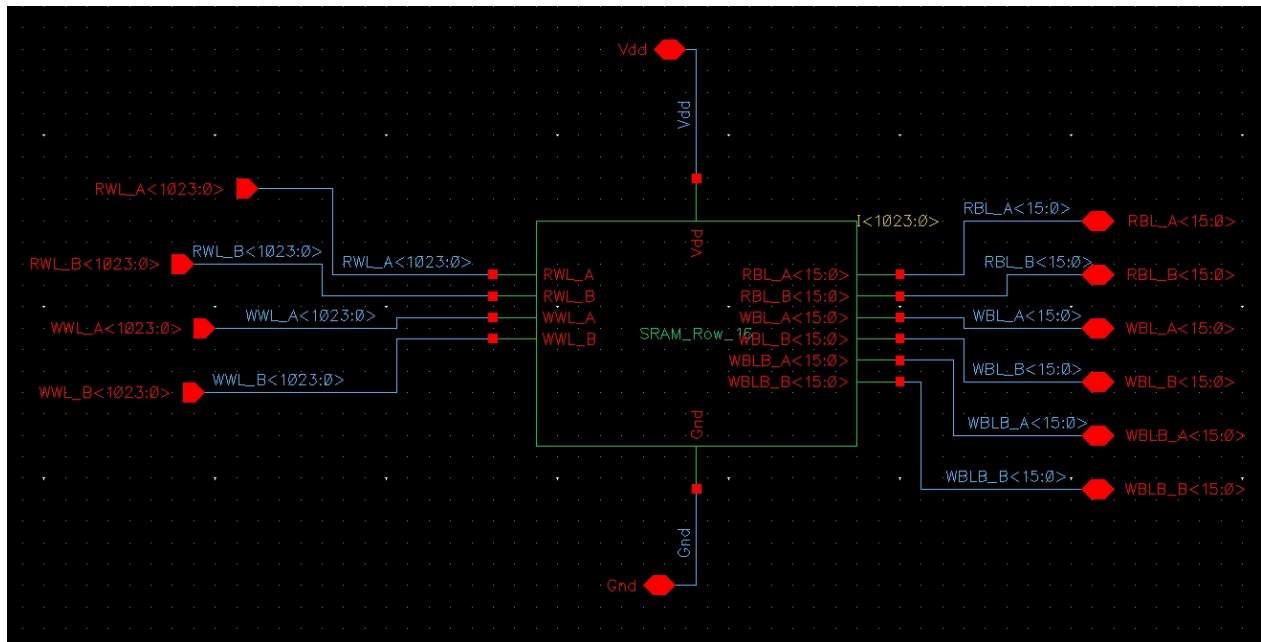


Fig5.5: 1024x16 Design

The Baseline SRAM_1024x16 array is implemented by instantiating 1024 rows of SRAM_Row_16, forming a complete 1024×16 bit memory array with a total capacity of 16,384 bits. The array uses a single continuous wordline spanning all 1024 rows, with shared bitlines across all columns for both read and write ports.

- **CLOCK GATE Cell Module**

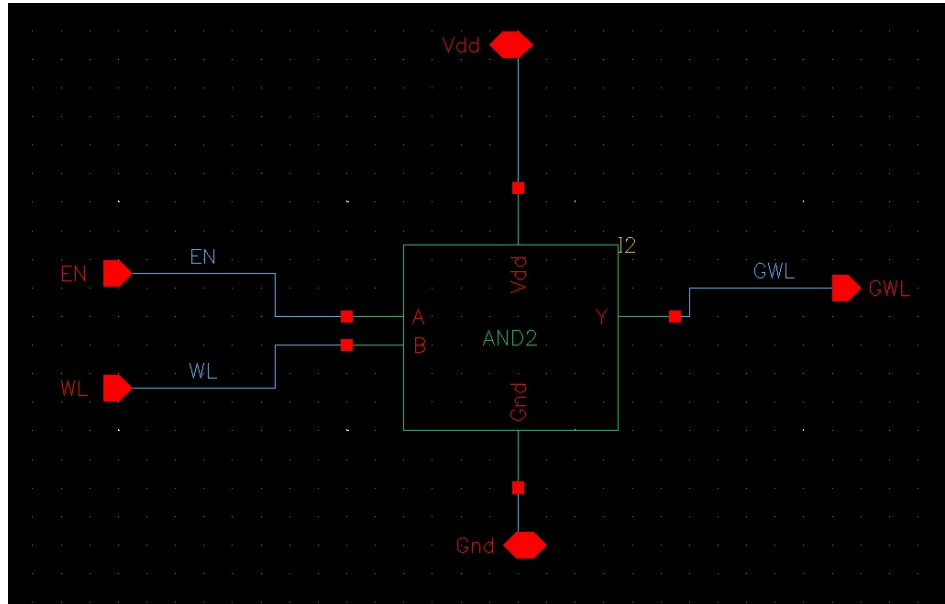


Fig 5.6:Clock Gate using 2 input And gate

The CLOCK GATE cell implements a two-input AND function: $GWL = EN \text{ AND } WL$. Sixteen instances of this cell are used in the optimized design four per segment, one for each wordline(WWL_A, WWL_B, RWL_A, and RWL_B). When the segment's EN signal (from SEG_EN<3:0>) is de-asserted, GWL remains at logic zero regardless of other inputs, completely preventing wordline switching in that segment

- **Optimized SRAM Module**

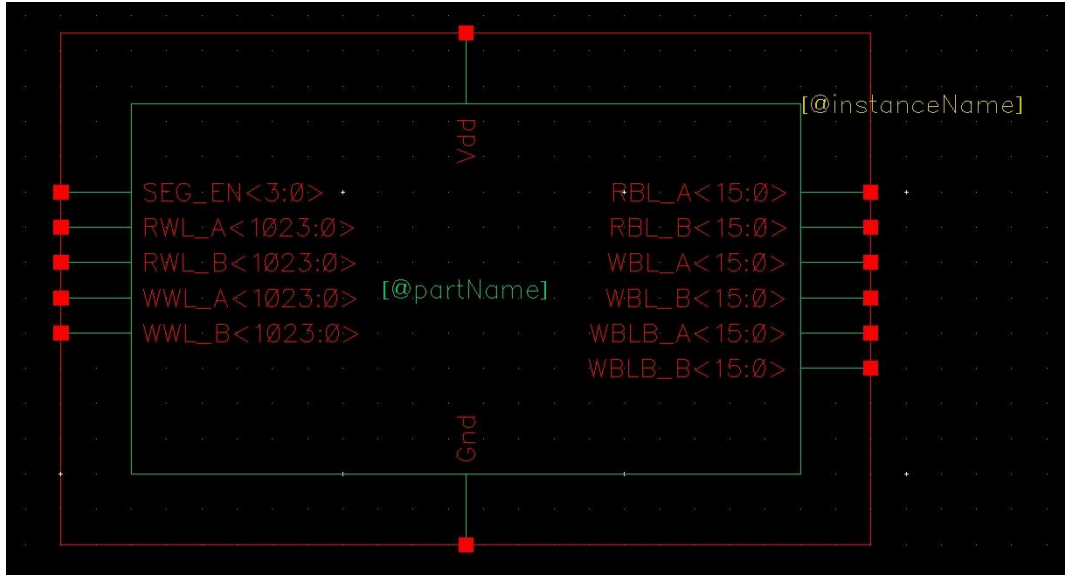


Fig 5.7 :Symbol of Optimized design of 1024x16 design using clock gating and wordline segmentation

The optimized top-level cell integrates all components: four SRAM_256x16 segment instances, sixteen CLOCK GATE instances, and the necessary signal routing. The top-level pins include the four wordline buses (WWL_A<1023:0>, WWL_B<1023:0>, RWL_A<1023:0>, RWL_B<1023:0>), six bitline buses, SEG_EN<3:0>, VDD, and GND. The CLOCK GATE outputs connect directly to the corresponding wordline inputs of the four SRAM_256x16 segments. All segments share the same bitline buses.

- **Testbench Module**

The testbench for the Baseline design drives all wordlines (WWL_A<1023:0>, WWL_B<1023:0>, RWL_A<1023:0>, RWL_B<1023:0>) simultaneously using vpulse sources, with all 1024 rows switching every cycle to represent the worst-case power scenario. For the optimized testbench, SEG_EN<0> = 1 active for 0-50ns, SEG_EN<1> = 1 active for 50-100ns, SEG_EN<2> = 1 active for 100-150ns, SEG_EN<3> = 1 active for 150-200ns limiting switching to only one segment at a time. Power is measured using the expression $\text{average}(1.2 \times \text{IT}(\text{VDD}))$ over the full 200 ns simulation.

5.3 Workflow

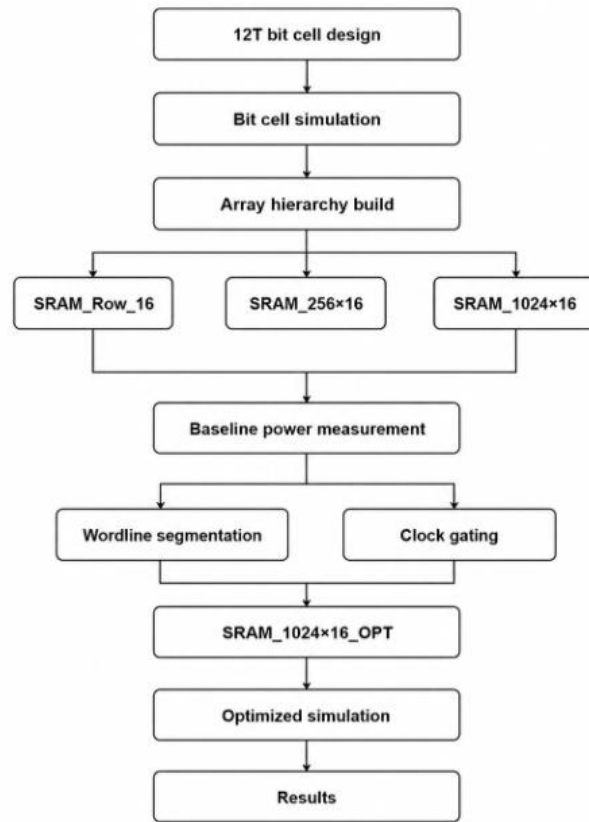


Fig 5.8 – Optimization Workflow

- **Design 12T Bit Cell Schematic:**

Create the 12T cell in Cadence Virtuoso using gpd090 NMOS and PMOS transistors. Connect storage core, write access transistors, and read path transistors according to the specified topology. Label internal nodes Q and QB as wire names (not pins). Verify transistor sizing and body connections.

- **Create Symbol and Simulate Bit Cell:**

Generate the symbol of the schematic. Build a testbench with vpulse sources for all wordlines and bitlines, and vdc with 10 kohms resistors for RBL precharge. Run 200ns transient simulation in ADEL to verify simultaneous Port A and Port B read/write operation from waveforms.

- **Build Array Hierarchy:**

Assemble SRAM_Row_16 by placing 16 bit cell instances and connecting shared wordlines with individual bitlines using bus notation. Scale up to SRAM_256x16 and then SRAM_1024x16 using Virtuoso's array instancing.

- **Measure Baseline Power:**

Create SRAM_1024x16_TB testbench with all wordline buses fully driven. Run 200 ns transient simulation. Use ADE L calculator expression $\text{average}(1.2 \times \text{IT}(\text{VDD}))$ to compute total average power. Record the Baseline power.

- **Design CLOCK GATE Cell:**

Implement the two-input AND gate with pins EN, WL (inputs) and GWL (output). Verify that $\text{GWL} = \text{EN AND WL}$ through simulation.

- **Design Wordline Segmentation:**

Divide the SRAM array into multiple segments (4 blocks). Use SEG_EN signals and the CLOCK GATE cell to activate only one segment at a time, while keeping others OFF. This reduces unnecessary switching and lowers dynamic power.

- **Assemble Optimized Design:**

Place four SRAM_256x16 instances and sixteen CLOCK GATE instances. Connect CLOCK GATE outputs to segment wordline inputs. Connect SEG_EN<3:0> and CLK pins. Share bitlines across all four segments.

- **Measure Optimized Power:**

Create optimized test bench with SEG_EN<0> = 1 active for 0-50ns, SEG_EN<1> = 1 active for 50-100ns, SEG_EN<2> = 1 active for 100-150ns, SEG_EN<3> = 1 active for 150-200ns. Run 200 ns transient simulation. Measure the power and calculate the reduction.

CHAPTER 6

RESULTS

6.1 Testbench Setup

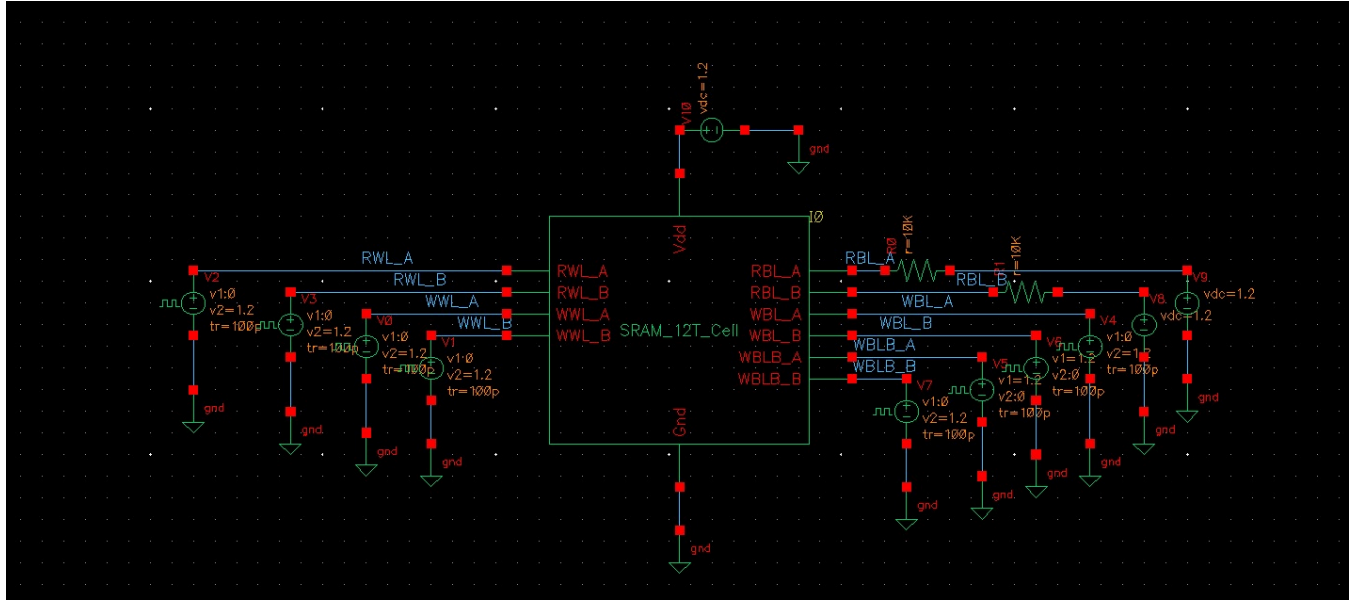


Fig 6.1: 12T Testbench

The testbench for the 12T SRAM cell was designed in Cadence Virtuoso using the gpd090 90 nm CMOS technology with a supply voltage of 1.2 V. Independent VPULSE sources were applied to the wordlines and bitlines of both Port A and Port B to validate simultaneous dual-port read and write operations under transient conditions. The timing of these pulses was carefully configured to ensure proper sequencing of read and write cycles for both ports without overlap or interference.

To accurately model the read operation, the read bitline (RBL) was precharged to VDD using a 10 kΩ resistor before each read cycle. This precharge mechanism enables correct sensing of voltage discharge during read access. The testbench setup allows verification of functional correctness, signal integrity, and stable data storage at internal nodes Q and QB under dual-port operation.

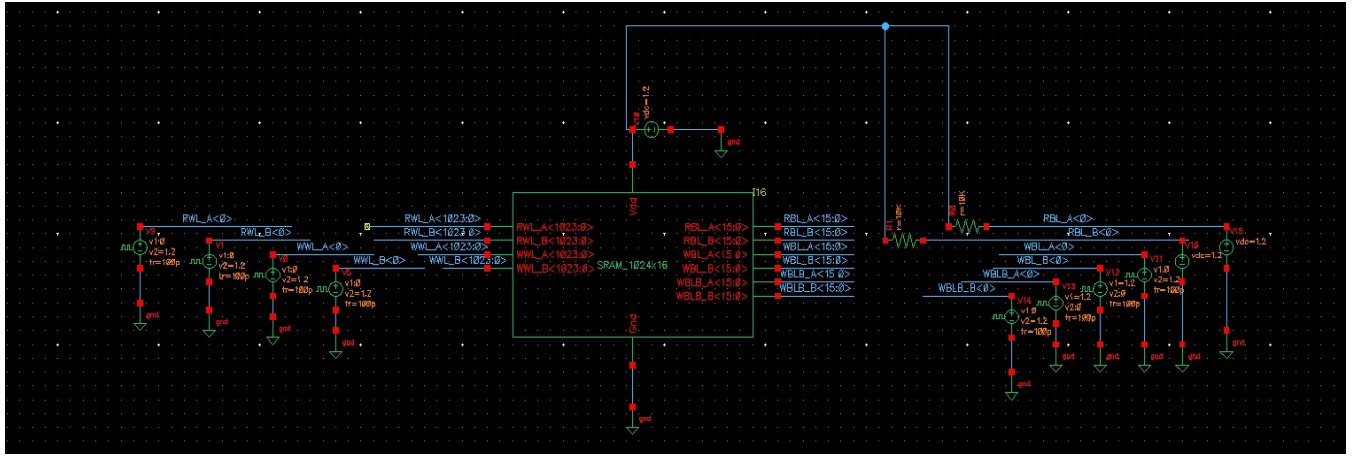


Fig 6.2:1024x16 Baseline testbench to calculate total power

The Baseline testbench for total power measurement was designed with vpulse sources driving both port A and port B wordlines and bitlines simultaneously to simulate active read and write operations. A 1pF decoupling capacitor was placed at the VDD node and 10kΩ precharge resistors were used for RBL nodes. Total power was extracted using the average current drawn from the VDD supply over a 200ns transient simulation.

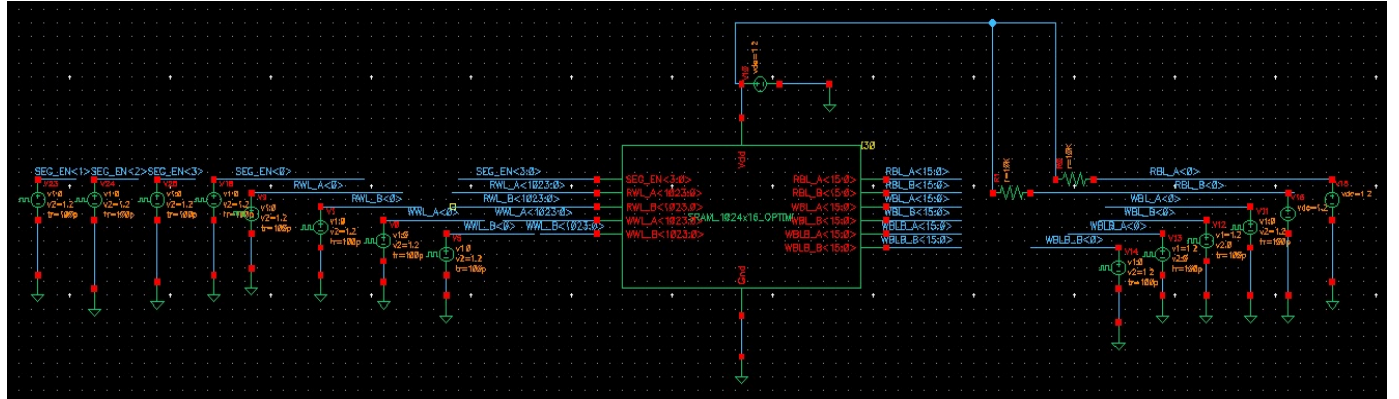


Fig 6.4: 1024x16 Optimized testbench to calculate total power

The optimized total power testbench drives the same vpulse signals as the Baseline testbench for active read and write operations, but only activates one segment at a time, while keeping wordlines of other segments at 0V. This configuration represents realistic operating conditions where only one segment is accessed per memory operation, and total power is measured using average VDD current over a 200ns transient simulation.

6.2 Waveform Analysis

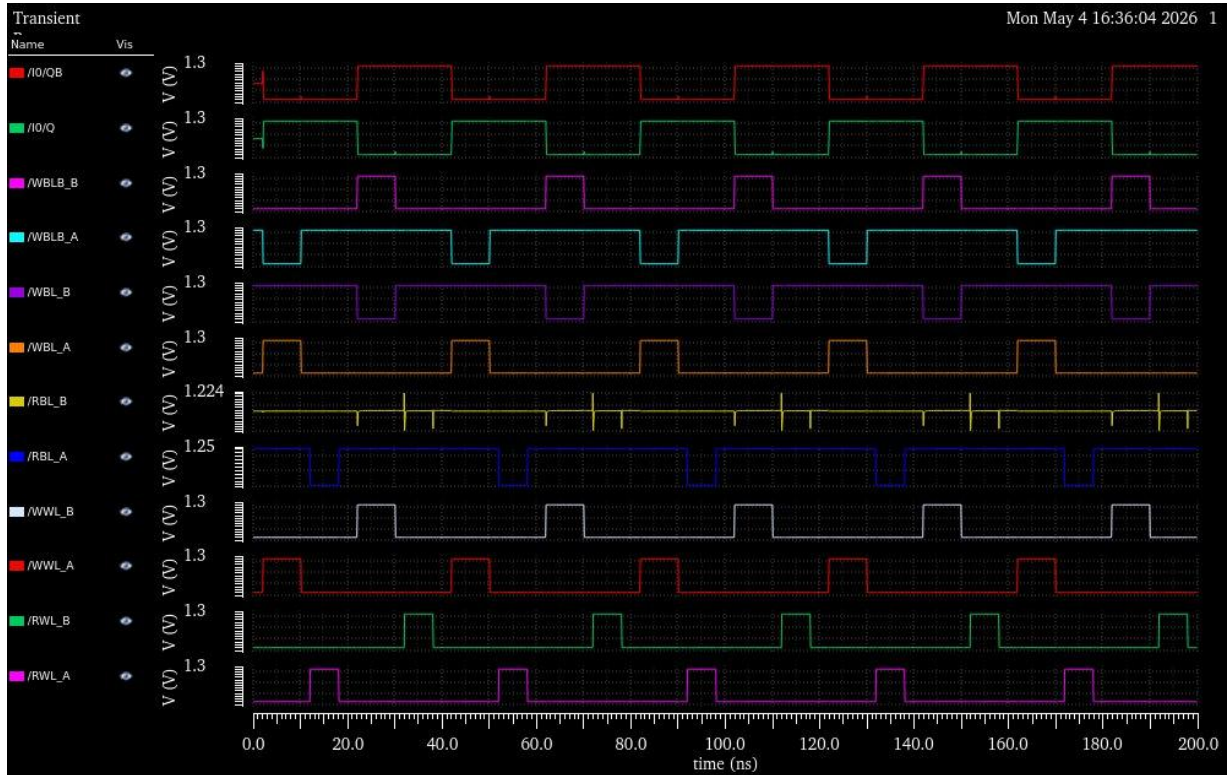


Fig 6.6:12T cell waveform

The transient waveform of the 12T SRAM cell demonstrates correct dual-port operation with clearly observable write and read cycles on both ports A and B. The storage nodes Q and QB show complementary switching behavior, confirming successful write operations, while the read bitlines respond correctly to the stored data during read operations without any disturbance to the storage nodes.

This also verifies the effectiveness of the decoupled read path, as the internal storage nodes remain stable throughout the read cycle. The absence of voltage fluctuation at Q and QB during read operation confirms disturbance-free sensing. Overall, the waveform validates reliable simultaneous operation of both ports with proper data integrity and stable cell behavior.

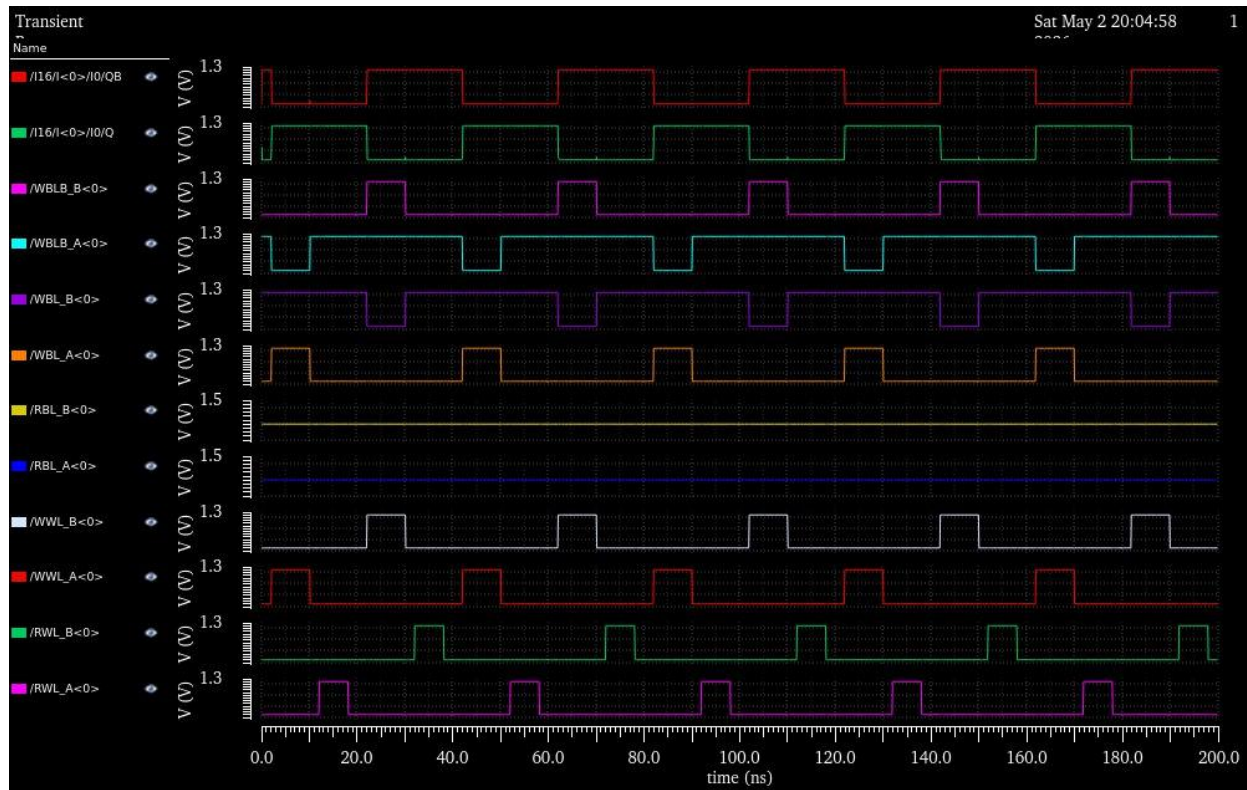


Fig 6.7:1024x16 Baseline waveform

The baseline transient simulation waveform shows correct operation of the 1024×16 SRAM array over a 200 ns simulation window. The write wordlines (WWL_A and WWL_B), read wordlines (RWL_A and RWL_B), and the corresponding bitlines toggle as per the applied input stimulus, confirming functional correctness of the complete array. The internal storage nodes Q and QB of the accessed cell exhibit proper complementary switching, verifying successful write and read operations.

This also indicates that the hierarchical design from the 12T cell to the full array is functioning reliably without any signal degradation. Continuous switching of all wordlines during the simulation reflects the worst-case operating condition, leading to high dynamic power consumption. The waveform therefore serves as a reference for baseline behavior before applying power optimization techniques.

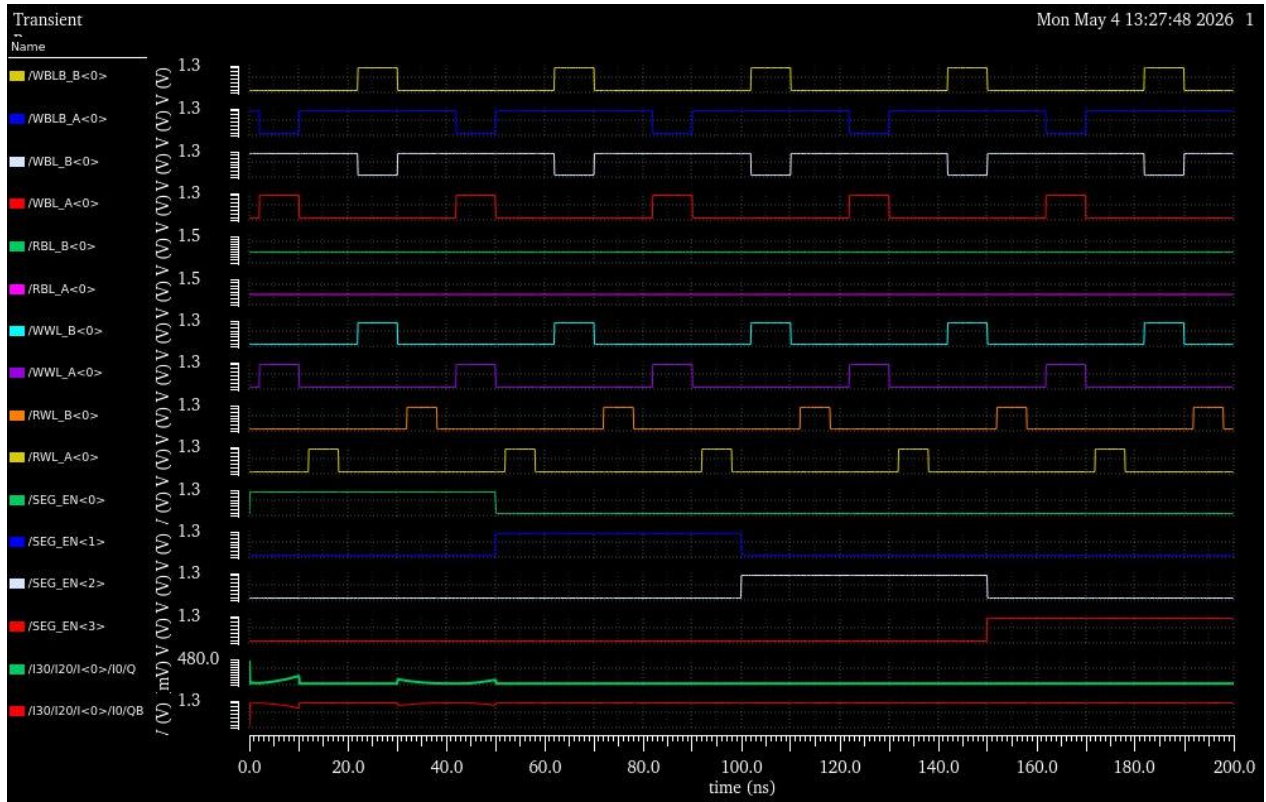


Fig 6.8:1024x16 Optimized waveform

The optimized transient waveform of the 1024×16 SRAM array confirms correct functional operation over the 200 ns simulation window. The segment enable signals (SEG_EN<3:0>) ensure that only one segment wordline is active at a time, while the remaining segment wordlines remain at 0 V, demonstrating effective wordline segmentation and clock gating.

The waveform appears slightly distorted compared to the baseline due to segmented activation, where switching occurs in different time intervals for each segment. Additionally, clock gating and reduced capacitive loading introduce variations in transition shapes, resulting in a non-uniform waveform. This behavior confirms reduced switching activity and improved power efficiency in the optimized design.

6.3 Power Analysis



Fig 6.9:Baseline power calculation

The baseline total power of the SRAM_1024x16 array was calculated and found to be 78.82mW using the Cadence Virtuoso calculator expression $\text{average}(1.2 \times \text{IT}("/\text{V10}/\text{MINUS}"))$ applied to the VDD supply current over the full 200ns transient simulation. The measured baseline power represents the reference power consumption of the unoptimized array under active read and write operations across both ports.

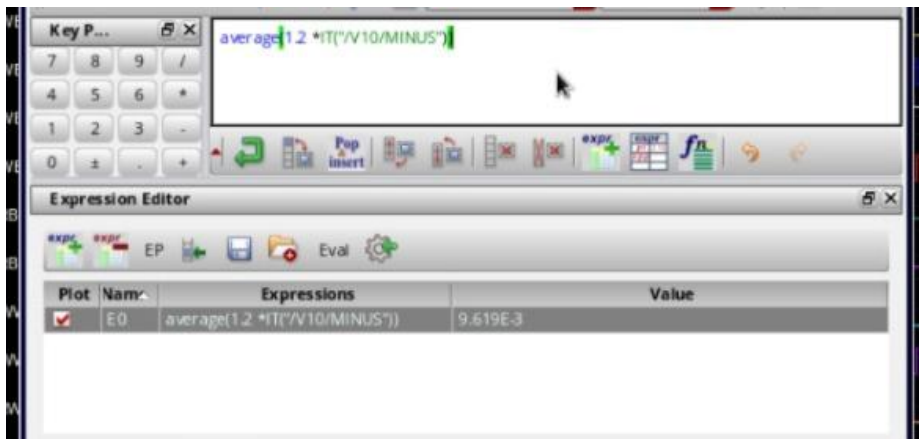


Fig 6.10:Optimized power calculation

The optimized power was calculated using the same calculator expression applied to the VDD supply of the SRAM_1024x16_OPT testbench under identical simulation conditions and found to be 9.619mW. The reduction in power consumption compared to the Baseline quantifies the effectiveness of the combined wordline segmentation technique, demonstrating the power saving achieved by restricting active switching to only one 256-row segment per memory access cycle.

CHAPTER 7

Conclusion and Future enhancements

7.1 Conclusion

The project "Design and Power Optimization of True Dual Port SRAM" successfully demonstrates the design, verification, and power optimization of a 12T True Dual Port SRAM in the gpdk090 90nm CMOS process using Cadence Virtuoso and Spectre.

The 12T cell topology provides two fully independent simultaneous read/write ports and a completely decoupled read path that eliminates read disturbance a fundamental limitation of conventional 6T and 8T SRAM cells. The full design hierarchy was implemented and verified through transient simulation at multiple levels single cell, 256×16 array and 1024×16 array.

Baseline power measurement confirmed 78.82 mW total average power under worst-case conditions, with all 1024 wordlines switching simultaneously at 25 MHz and 1.2V supply. Two power optimization techniques wordline segmentation and clock gating were applied to the complete array. Wordline segmentation divided the 1024-row array into four independent 256-row segments.

Clock gating using 2 input AND -based CLOCK GATE cells blocked wordline switching in the three inactive segments by holding their gated wordlines permanently at logic zero. With only one segment active at a time, the optimized design measured 9.619 mW a reduction of approximately 92% compared to the Baseline.

These results confirm that the combination of wordline segmentation and clock gating is a highly effective strategy for reducing dynamic power in large SRAM arrays. The design is fully functional and validated in an industry-standard EDA environment, providing a strong foundation for extension to physical implementation and advanced optimization.

7.2 Future Enhancements

- **Sense Amplifier Integration:** Design and integrate latch-type sense amplifiers on each column's read bitline to detect small RBL discharge voltages, enabling faster and more reliable read operations at reduced supply voltages.
- **Row Decoder Design:** Implement a 10 to 1024 NAND-based row decoder to enable address-driven wordline selection, transforming the design into a fully functional addressed memory array.
- **Corner Analysis:** Conduct post-layout simulations across all five PVT corners (TT, FF, SS, FS, SF) and over the temperature range from 40°C to 125°C to verify power and timing margins under worst-case conditions.
- **Larger Array Extensions:** Scale the design to 1024×64 and 4096×16 configurations to quantify how wordline segmentation and clock gating benefits scale with array dimensions, validating the architectural approach for production-scale memories.

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Appendix

Development Cost

Component / Tool	Description	Nos	Cost
Cadence Virtuoso IC617	Schematic entry and simulation environment	1 (license)	Academic license
gpd090 PDK	90nm CMOS process design kit	1	Academic license
Spectre Simulator	Transient and DC analysis engine	1 (bundled)	Included with Virtuoso
Linux Workstation	RHEL6 / CentOS with 16GB RAM, 8-core CPU	1	Lab provided
Cadence ADE L	Analog Design Environment for simulation control	1 (bundled)	Included with Virtuoso

Table1: Development Cost

Simulation Results Summary

Parameter	Baseline	Optimized	Reduction
Technology	gpd090 (90nm)	gpd090 (90nm)	—
Supply Voltage (VDD)	1.2V	1.2V	—
Clock Frequency	25 MHz	25 MHz	—
Array Size	1024 × 16	1024 × 16	—
Active Rows	1024 / 1024 (all)	~256 / 1024	75% rows gated
Total Average Power	78.82 mW	9.619 mW	87%
Power Saved	—	69.201 mW saved	—
Simulation Duration	200 ns transient	200 ns transient	—
Power Formula	average($1.2 \times IT(VDD)$)	average($1.2 \times IT(VDD)$)	—

Table 2: Simulation Results Summary